Nuclear Engineering and Design 50 (1978) 411–424

REVIEW OF TIME-DEPENDENT FATIGUE BEHAVIOR OF STRUCTURAL ALLOYS * **

W.L. GREENSTREET
Oak Ridge National Laboratory, Oak Ridge, Tennessee 37830, USA

Received 24 April 1978

"A review and assessment of time-dependent fatigue was needed to provide an understanding of time-dependent fatigue processes, to define the limits of our present knowledge, and to establish bases for the development of verified design methods for structural components and systems for operation at elevated temperatures. This report reviews the present state of understanding of that phenomenon, commonly called 'creep fatigue', and separates it into crack-initiation and crack-propagation processes. Criteria for describing material behavior for each of these processes are discussed and described within the extent of present knowledge, which is limited largely to experience with one-dimensional loading. Behaviors of types 304 and 316 stainless steel are emphasized. Much of the treatment of time-dependent failure presented here is new and of a developing nature; areas of agreement and areas requiring further resolution are enumerated."

These words are from the abstract of the report on a comprehensive study of time-dependent fatigue. This paper briefly reviews some of the contents and discusses important conclusions reached, especially in terms of current status and needs for additional work.

1. Introduction

The results of a comprehensive study [1] of time-dependent fatigue behavior of structural alloys at elevated temperatures is described. This study was undertaken by a group of preeminent individuals to summarize and assess the current state of knowledge and to define directions for future research and development work. Structural behavior evaluation and design code rule development requirements were considered together with correlative and predictive capabilities in making the assessments and in establishing requirements for future progress.

Phenomenological approaches were considered and, to treat the fatigue failure process as a whole, both crack-initiation (CI) and crack-propagation (CP) aspects were addressed. The investigators used frameworks that were familiar to them to bring out salient features of behavior observed, to give perspective and coherence to these observations, and to provide logical descriptions of the phenomena involved. Also, for perspective and continuity, attempts were made to define boundaries between CI and CP stages.

The time-dependent fatigue designation was adopted in recognition that many time-dependent processes, including chemical and metallurgical changes, occur and combine with inelastic strain reversals to produce fatigue damage and failure. Because of the many basic factors and the complexities of behavior involved, emphasis was given to the study of uniaxial responses of materials under repeated cycle conditions. Development of correlative and predictive capabilities for such cases is a prerequisite to successful treatment of more complex conditions.

To provide a design application focus and to delineate key aspects of interest to design analysts, an initial chapter of ref. [1] was addressed to analysis applications and structural design code requirements. However, the contents of this chapter will not be summarized here.

The succeeding sections address crack-initiation and crack-propagation phases, followed by brief considerations of the overall fatigue process as portrayed by

* Research sponsored by the Division of Reactor Research and Technology, US Department of Energy under contract W-7405-eng-26 with the Union Carbide Corporation.
** Expanded version of Invited Paper L 9/1 presented at the 4th International Conference on Structural Mechanics in Reactor Technology, San Francisco, California, 15–19 August 1977.

results from low-cycle-fatigue specimens. The discussions center primarily on particular methods for correlation and prediction since these methods reflect summaries and interpretations of the complex behaviors observed and allow development of a coherent assessment of knowledge; this course facilitates identification of areas of strength and weakness and permits succinct delineations of the areas requiring additional study. This approach corresponds to that used in the study report.

2. Fatigue failure process

The overall fatigue failure process can be depicted as shown in fig. 1. Three phases, or stages, are involved: crack initiation, crack propagation and an instability phase which is associated with tearing and/or rapid crack growth leading to terminal fracture. Although each phase is important, the first two are of primary interest because generally the useful life of a structure or component ends when the third stage is reached. The crack-initiation phase, as discussed in the report [1], includes both microcrack initiation and early macrocrack growth. At the end of this phase a crack of very small but finite size exists; for example, the size of the crack may be 1 to 3 grain dimensions for a relatively fine-grained structural alloy. The duration of the initiation phase is influenced by a number of variables; the roles of inelastic strain, wave shape, temperature, strain rate, and the many other variables involved are topics of the report.

To illustrate the relative importance of the initiation phase, durations that can be expected for type

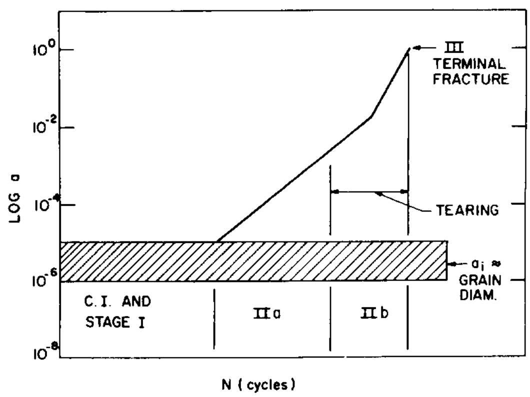


Fig. 1. Stages of fatigue failure process.

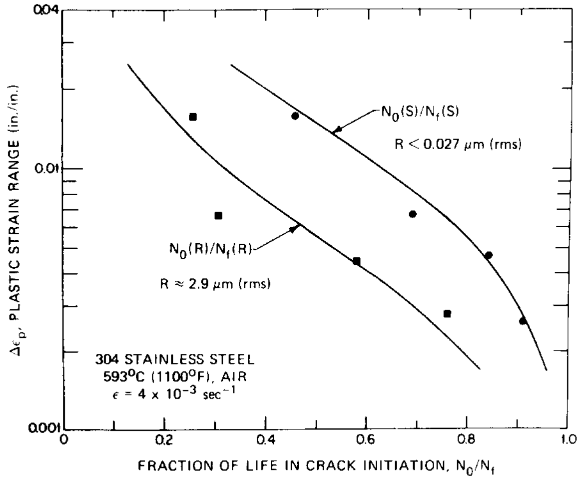


Fig. 2. Effect of plastic strain range on the fracture life spent in crack initiation.

304 stainless steel at 593°C (1100°F) in air are shown in fig. 2 from Maiya [2], where the ratio of cycles to initiation (N_0) to cycles to failure (N_f) for low-cycle-fatigue specimens is given as a function of strain range. Here initiation corresponds to a crack length of about 0.2 mm (0.008 in.), and surface roughness is taken as a parameter. The figure shows that for strain ranges experienced in many applications (0.2 to 0.5%), the crack-initiation phase may account for a large percentage of total life. It also indicates that surface roughness can be of crucial importance.

Crack extension in the second phase involves macrocrack extension as well as growth of larger cracks. Fracture mechanics concepts and methods apply to behavior in this stage. Precise demarcations between the initiation and propagation stages and lower limits on fracture mechanics applicability are yet to be established. Also to be defined quantitatively is the terminus of the second phase; however, several candidates are suggested in this case.

3. Crack initiation

An extensive amount of work has been directed toward understanding the crack-initiation phase and toward deriving methods for correlating and predicting the associated behavioral patterns. Two methods for describing the behavior are to be considered in this case:

the frequency-separation life relation method, due to Coffin [3], and the strainrange partitioning method of Manson and coworkers [4,5]. The discussions given in ref. [1] regarding these methods refer to the generation of a crack about 2.5 mm (0.1 in.) long, and the experimental observations made to determine the constants are consistent with this definition. The objective of the development in each case was to provide detailed descriptions of key features of time-dependent fatigue behavior in a generalized way. The resulting methods provide excellent vehicles for gaining understanding and for further study of the multifarious nature of the time-dependent fatigue process.

The first method was introduced during the course of the assessment project, while the second was extended and embodied concepts were examined in greater detail. Both are based on extensions of methods developed by Coffin and Manson for treating low-cycle fatigue at low temperature.

The point of departure in each case is the well-known Coffin–Manson equation for low-cycle fatigue,

$$\Delta\epsilon_p = AN_f^{-\beta} . \tag{1}$$

This is combined with the Basquin equation,

$$\Delta\epsilon_e = A'N_f^{-\beta'} , \tag{2}$$

to give the total strain range equation

$$\Delta\epsilon = \Delta\epsilon_p + \Delta\epsilon_e = AN_f^{-\beta} + A'N_f^{-\beta'} , \tag{3}$$

where

$\Delta\epsilon_p$ = inelastic strain range,
$\Delta\epsilon_e$ = elastic strain range,
N_f = cycles to failure,
A, β, A', β' = material constants.

3.1. Frequency-modified and frequency-separation life relations

The frequency-modified life relations were derived by Coffin [6,7] from the above equation by including the time per cycle (t) explicitly. This was done on the premise that time must be introduced because of environmental and other time-dependent influences. Cycle frequency, ν (= $1/t$), was used because environmental damage depends on cycle period. Thus, by introducing frequency, the above equations become

$$\Delta\epsilon_p = A_1(N_f\nu^{k-1})^{-\beta} , \tag{1a}$$

$$\Delta\epsilon_e = A_1'N_f^{-\beta'}\nu^{k_1'} , \tag{2a}$$

$$\Delta\epsilon = A_1(N_f\nu^{k-1})^{-\beta} + A_1'N_f^{-\beta'}\nu^{k_1'} . \tag{3a}$$

In these equations, A_1, k, β, A_1', k_1' and β' are temperature-dependent constants; the frequency-modified lives are $N_f\nu^{k-1}$ in eq. (1a) and $N_f\nu^b$ in eq. (2a). The second life term can be obtained from eq. (2a) after some manipulation; b is again a constant.

Recognizing that wave shape as well as time is important, additional modifications were made. It was postulated that damage due to wave shape influences is associated with tension-going * time and hysteresis loop imbalance. The equations which resulted are the frequency-separation life relations. To obtain the inelastic strain expression in this case, the frequency-modified equation, eq. (1a), is written as

$$N_f = \left(\frac{A_1}{\Delta\epsilon_p}\right)^{1/\beta} \nu^{1-k} . \tag{1b}$$

This expression is then altered to give the frequency-separation equation

$$N_f = \left(\frac{A_1}{\Delta\epsilon_p'}\right)^{1/\beta} \left(\frac{\nu_t}{2}\right)^{1-k} , \tag{1c}$$

where $\Delta\epsilon_p'$ is equivalent plastic strain,

$$\Delta\epsilon_p' = \Delta\epsilon_p\left[\frac{(\nu_c/\nu_t)^{k_1} + 1}{2}\right]^{\beta/\beta'} , \tag{4}$$

and

$$k_1 = k_1' + \beta'(k - 1) . \tag{5}$$

Here $\nu_t = 1/t_t$ = tension-going frequency and $\nu_c = 1/t_c$ = compression-going frequency. The total frequency is given by

$$\nu = \frac{1}{1/\nu_t + 1/\nu_c} . \tag{6}$$

In eq. (1c), $(\nu_t/2)^{1-k}$ accounts for tension-going time and $\Delta\epsilon_p'$ accounts for loop imbalance.

A number of ramifications, including treatment of complex wave forms, are discussed in ref. [1]. Since the method is still in the early stages of development, its potential is largely unexplored.

The frequency-separation life relation method, or

* Tension-going refers to that part of the hysteresis loop in which the plastic strain rate is positive.

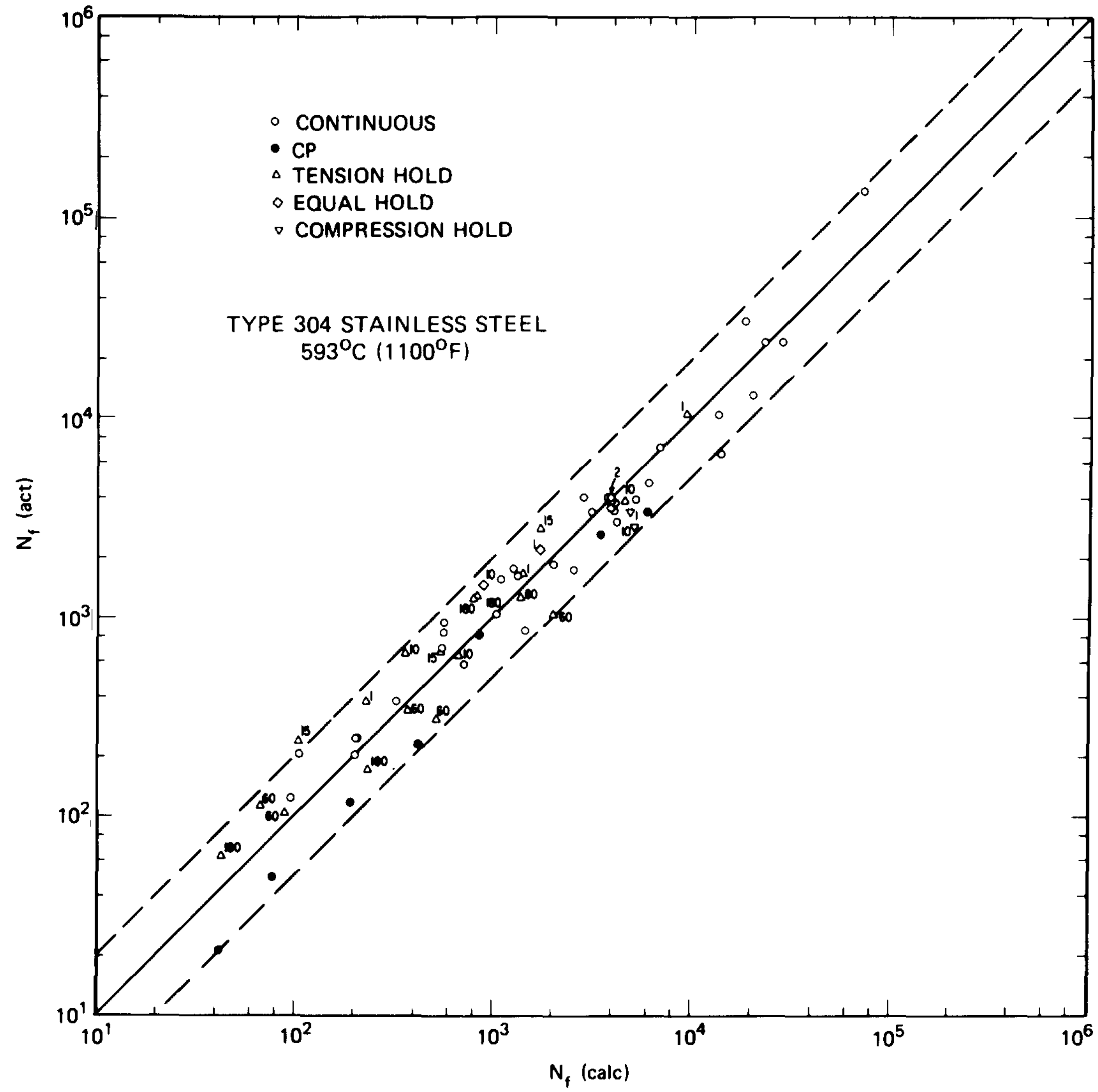


Fig. 3. Comparison of ANL experimental results and frequency separation predictions.

simply the frequency-separation method, gives good correlations between observed and predicted lives, as shown in fig. 3. The experimental data were obtained at the Argonne National Laboratory (ANL) [8a,b] from type 304 stainless steel specimens for various constant strain ranges, except for the CP results which were obtained by stress cycling (cyclic creep). The wave shapes are listed in the figure, and the hold times (in minutes) are given beside the appropriate data points. The maximum hold times were 180 min for tension hold, 10 min for equal tension and compression hold, and 10 min for compression hold. Essentially all results fall within a factor of 2 scatter band.

There are several important general observations to be made concerning the frequency-separation approach. It is obvious that strain is a major correlating variable, but inelastic strain is not subdivided. Further division of the strain is not made because it was concluded that inelasticity is associated with a whole spectrum of flow mechanisms that are too complex to separate. Time is important because of environmental interactions and other time-dependent processes. Wave shape is also important, especially in terms of tension-going time and loop imbalance, i.e. the unbalanced or mean stress associated with the hysteresis loop. Temperature and strain rate are two basic independent variables. Environmental and other time-dependent effects are given cognizance through the frequency terms. Implicit in the method is an assumption that the significant damage for a cycle occurs near the peak tensile strain and is affected by the strain rate in approaching this extremum.

More specifically, total strains are addressed, and the equations were derived to treat low strains. However, large strains can be treated as well. The equations predict fatigue lives which decrease indefinitly with decreasing frequency for constant temperature, inelastic strain range, and wave form. In addition, progressive changes in life occur with changes in temperature and frequency ratio (ν_c/ν_t).

3.2. Strainrange partitioning method

The underlying bases for the strainrange partitioning method are as follows.

(1) Two modes of inelastic deformation – plastic flow and creep – may occur separately or concurrently, and their interaction can strongly influence fracture behavior.

(2) The manner in which a tensile component of inelastic strain is balanced by a compressive component to close a hysteresis loop during completely reversed straining is to be featured by a predictive method.

On the basis of these premises, four basic cases of "pure" inelastic strain component combinations, or "generic strain ranges", can be identified. It is further postulated that the material integrates the time-dependent influences to provide unique basic life relationships and the fatigue lives can be determined on the basis of these relationships. The life relationships are essentially temperature independent; however, it is recognized that metallurgical instabilities which affect ductility can negate the temperature-independent assumption.

The four basic types of hysteresis loops are shown in fig. 4; the first subscript used in a strain range designation refers to the type of tensile strain, while the second refers to the type of compressive strain. The four generic strain ranges are shown in table 1. Only three of these strain ranges can exist in a given hysteresis loop; a simple example of mixed strain ranges is shown in fig. 5.

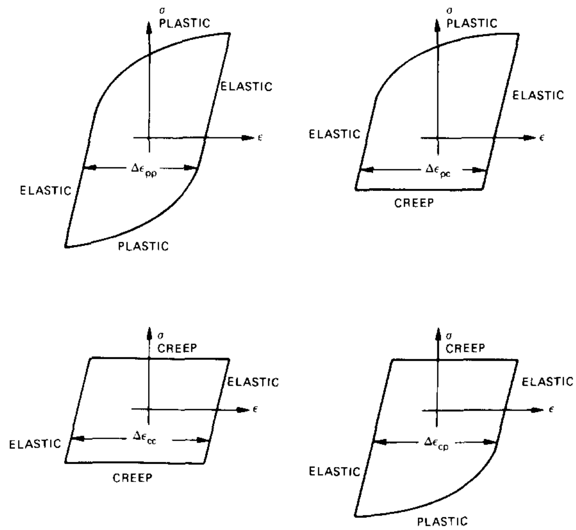


Fig. 4. Idealized hysteresis loops for the four basic types of inelastic strain range.

The basic life relationships used are given by Manson–Coffin-type equations:

$$\Delta\epsilon_{cp} = AN_{cp}^{-\alpha}, \quad (7)$$

$$\Delta\epsilon_{pc} = BN_{pc}^{-\beta}, \quad \text{etc.}, \quad (8)$$

where A, B, α, β, etc. are material constants. Universalized life relations are obtained by expressing the coefficients A, B, etc. in terms of ductilities. The tensile ductility (D_p) is used when the tensile component of strain is plastic, while creep ductility (D_c) is used when the tensile component is creep strain. The basic life relations are functions of material, as shown in fig. 6. For type 304 stainless steel, fig. 6 shows that the $\Delta\epsilon_{cp}$ strain gives the lowest life, i.e. tensile hold is most damaging. The life relationships for type 316 stainless steel are similar to those for type 304. How-

Table 1

Strain range	Tensile strain	Compressive strain
$\Delta\epsilon_{pp}$	plastic	plastic
$\Delta\epsilon_{pc}$	plastic	creep
$\Delta\epsilon_{cc}$	creep	creep
$\Delta\epsilon_{cp}$	creep	plastic

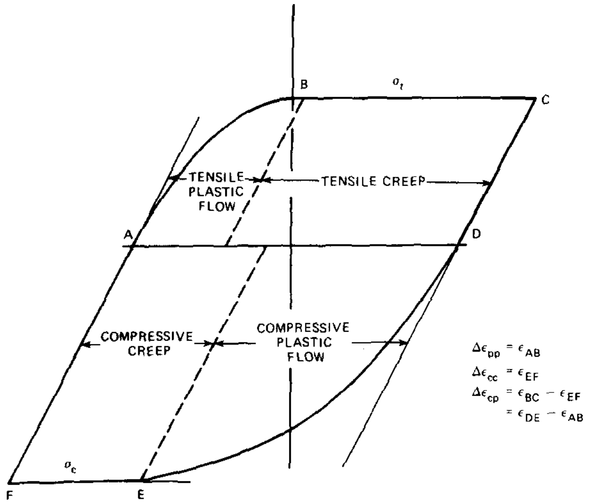


Fig. 5. Hysteresis loop producing strain range components, e_{pp}, e_{cc} and e_{cp}.

ever, the results for $2\frac{1}{4}$ Cr–1 Mo steel are almost totally different. For strain ranges less than 1%, compressive hold is the most damaging, but for ranges greater than 1%, tensile hold produces the greatest damage. At 1%, $\Delta\epsilon_{pc}$, $\Delta\epsilon_{cc}$ and $\Delta\epsilon_{cp}$ are equally damaging.

Cumulative damage from cycle to cycle is assumed to be linear, but an "interaction damage rule" is used to determine the damage per cycle. According to this rule, the damage per cycle is given by

$$\frac{f_{pp}}{N_{pp}} + \frac{f_{pc}}{N_{pc}} + \frac{f_{cc}}{N_{cc}} + \frac{f_{cp}}{N_{cp}} = \frac{1}{N_f} . \tag{9}$$

The f values are strain range fractions, i.e.

$$f_{pp} = \Delta\epsilon_{pp}/\Delta\epsilon , \quad \text{etc.,} \tag{10}$$

where

$$\Delta\epsilon = \Delta\epsilon_{pp} + \Delta\epsilon_{pc} + \Delta\epsilon_{cc} + \Delta\epsilon_{cp} ,$$

and N_{pp}, N_{pc}, etc. refer to points on life lines corresponding to $\Delta\epsilon$. An example is shown in fig. 7 for a combination of $\Delta\epsilon_{cc}$ and $\Delta\epsilon_{pp}$ strains. The total inelastic strain is 1%, and the number of cycles corresponding to points A and B are N_{cc} and N_{pp}, respectively. The damage per cycle is then calculated on the basis of the remaining data given.

The strainrange partitioning method, as in the case of the frequency-separation method, gives good agree-

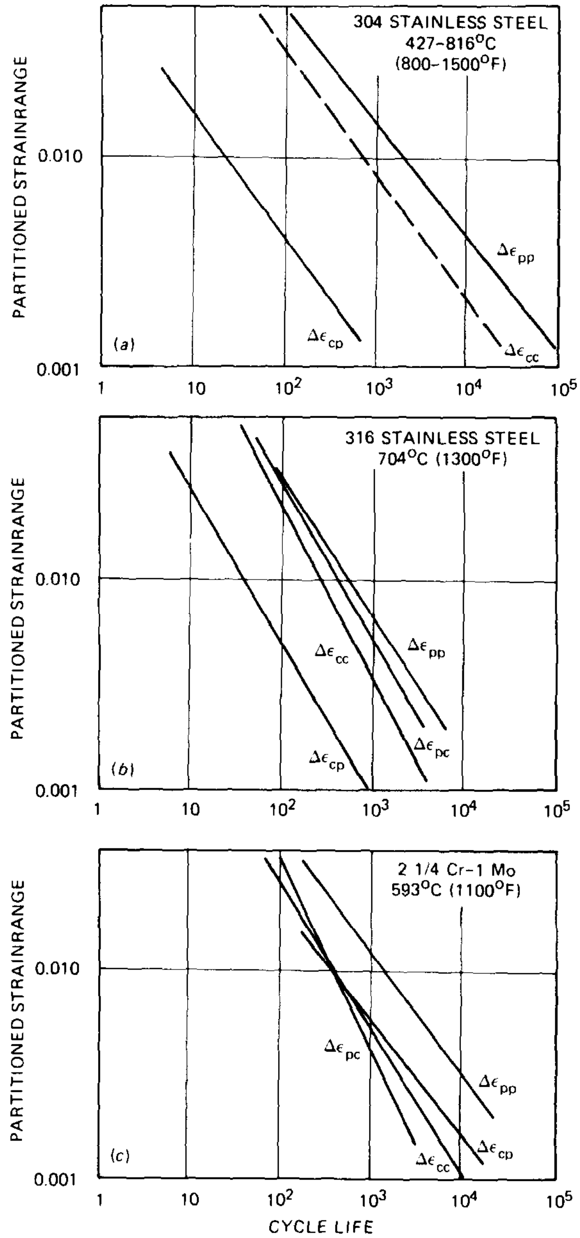


Fig. 6. Combined strain range partitioning relations.

ment between observed and calculated results, as shown in fig. 8. The data are for type 316 stainless steel and are based on results obtained by Kanazawa and Yoshida [9] (upper) and by Udoguchi et al. [10] (lower). In the upper portion of the figure the strain rate is the major variable, while in the lower portion

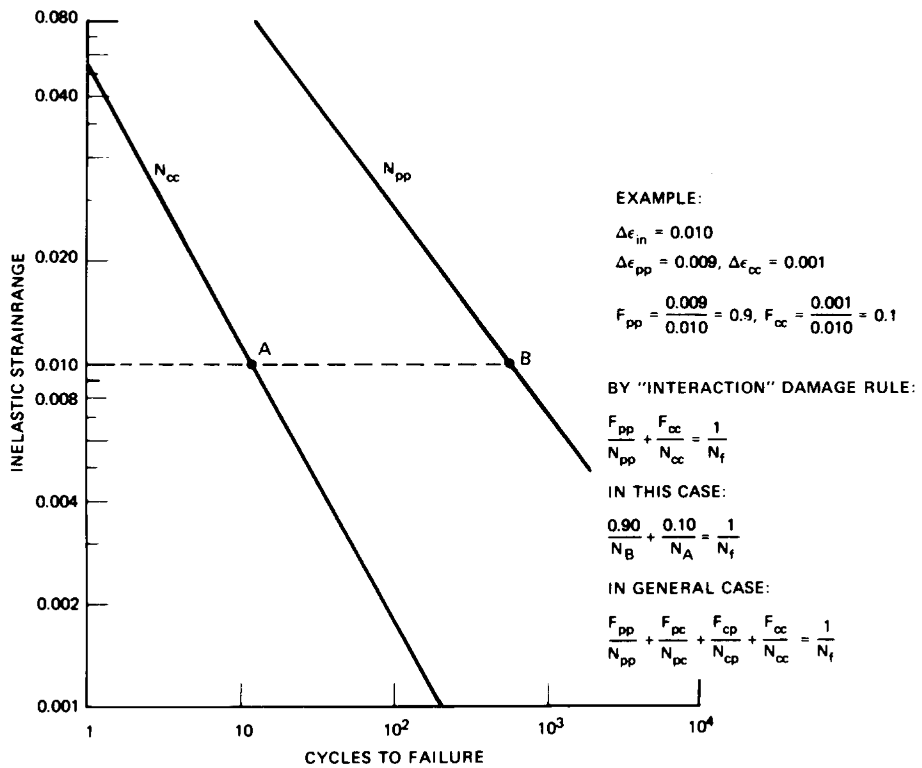
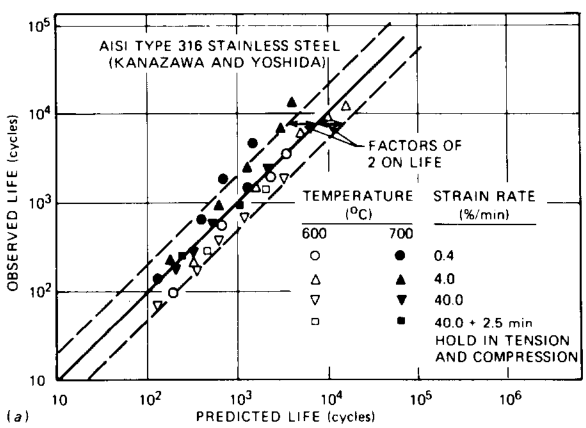


Fig. 7. Relationships for determining life.

the hold time is the parameter of interest. Different temperatures are considered in each case. The overall agreement is good, with a greater tendency for some of the higher temperature data to fall outside the factor of 2 band.

General observations to be made concerning the strainrange partitioning method parallel those for the frequency-separation method. Again, the major correlating variable is strain; however, in this case, the inelastic strain is subdivided into plastic and creep com-



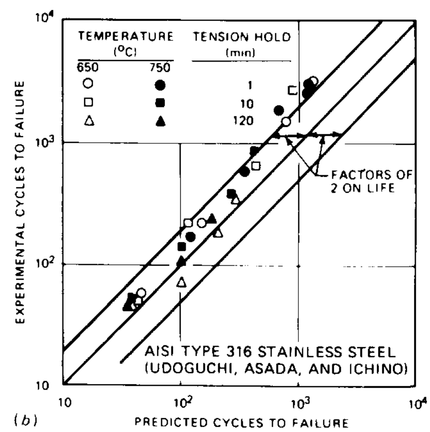


Fig. 8. Comparisons of experimental results and predictions based on strain range partitioning.

ponents. Only inelastic strains are considered. Time is accounted for since the strain rate affects the type of deformation incurred and the ductility. Wave shape is an important factor and is treated directly. Temperature influences the partitioned strains, and environmental effects are introduced through the basic life relations. However, only those environmental effects that occur concurrently with straining are treated.

Specific aspects especially noteworthy in this case are as follows. The basic postulates and treatment of inelastic strain cycling are compatible with curretly employed constitutive equation descriptions of material behavior. A major asset is the capability provided for examining and characterizing hold-time effects; for example, relative influences of hold times in tension and compression can be clearly shown. Another attractive feature is the relative insensitivity of the life relations to temperature when compared with other methods. Unlike frequency-separation method predictions, the strainrange partitioning method implies that there is a lower limit on life with decreasing frequency for a given inelastic strain range. Finally, the basic method is applicable primarily to large inelastic strains. In a more recent study [11], the method was extended to treat small strains.

As stated at the outset, both methods provide excellent vehicles for studying crack-initiation behavior. However, both apply to repeated cycle loading histories in which all cycles are identical. Linear summation of cycle fractions is used in each case for cumulative damage determinations. The implementation of these methods in design evaluations is a separate matter and will not be discussed here.

For multiaxial stress states, the use of effective stress range and strain range quantities is proposed by Coffin. Effective, or equivalent, stress and strain quantities are to be used in conjunction with the strainrange partitioning method. For the latter, the equivalent stress stress is defined in terms of principal stresses by

$$\sigma_e = \tfrac{1}{2}\sqrt{2}[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2]^{1/2} , \tag{11}$$

and the equivalent strain is defined in terms of principal strains by

$$\epsilon_e = \tfrac{1}{3}\sqrt{2}[(\epsilon_1 - \epsilon_2)^2 + (\epsilon_2 - \epsilon_3)^2 + (\epsilon_3 - \epsilon_1)^2]^{1/2} . \tag{12}$$

For discussion, it is assumed here that the elastic strains are negligible in comparison with the inelastic strains. The definitions establish the equivalent stresses and strains as positive definite scalars. No cognizance is given to directionality.

For analyses using the strainrange partitioning concept, hysteresis loops are constructed using equivalent stress and strain and the loops are decomposed as in the uniaxial case. The equivalent stresses and strains are assigned algebraic signs corresponding to the signs of stress and strain, at any instant of loading, for the direction which gives the lowest life prediction. The sign donor in many cases will be from the principal stress direction in which the largest stress change occurs, but the possibility of other directions giving lower predicted lives must be checked.

A number of ramifications of the treatment of multiaxial cases are discussed in the report [1], but the added features involved will not be covered here. A major difference from past practice in handling multiaxial cases is a rudimentary recognition that both direction and magnitude for the correlating variables are important.

4. Fatigue crack growth

The crack-initiation phase, or stage I, ends at some number of cycles (N_0) after which the crack-growth stage (stage II) dominates, as shown in fig. 9. Although N_0 and the associated crack size lack precise definition, this deficiency does not detract from the concepts being pursued. Three basic modes of crack surface displacement are to be considered in stage II; these are shown in fig. 10. However, attention has been focused primarily on mode I, and various data correlation methods have been used for this mode.

For elevated-temperature mode I, stage II fatigue crack growth, correlations have been made for two basic cases: low strain and large crack size and large strain and small crack size. For the first case, da/dN, or the increment of crack extension, da, per cycle, N, is correlatable as a function of ΔK, the range of stress intensity. Other parameters, such as the effective stress-intensity factor K_e and the quantity $(K_e^2 - K_{eth}^2)$, where K_{eth} is the threshold K below which a crack will not grow, are also used. The effective stress intensity factor is given by

$$K_e = K_{max}(1 - R)^m , \tag{13}$$

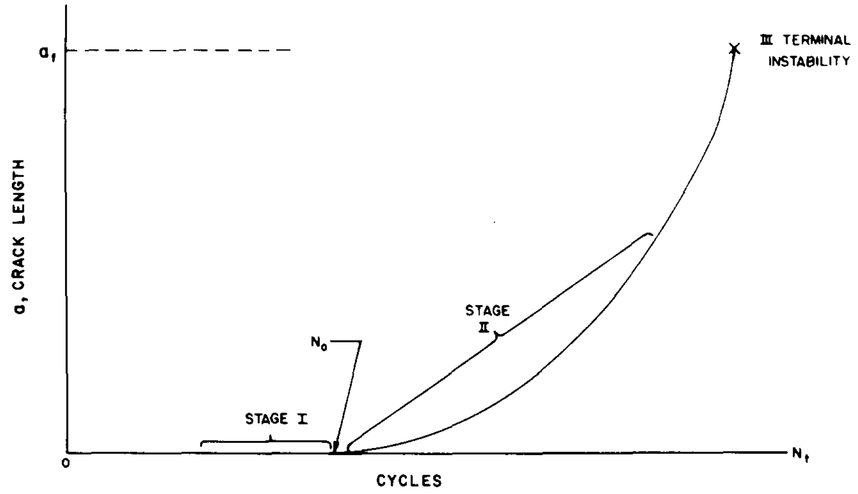


Fig. 9. Fatigue crack-growth and crack-initiation phases in the failure process.

where m typically equals 1/2, and

$$R = K_{min}/K_{max} \, .$$

The quantity K_{max} is the maximum stress intensity and K_{min} is the minimum. The term $K_e^2 - K_{eth}^2$ can be expressed in terms of stress intensities by

$$K_e^2 - K_{eth}^2 = (K_{max}^2 - K_{th}^2)(1 - R)^{2n} \, , \tag{14}$$

where K_{th} is the threshold stress intensity and n is a constant.

For large strains and small cracks, the crack length (a) can be correlated as a function of the number of cycles N. A second method is to plot da/dN as a function of a. Linear relationships are obtained or can be derived in some cases. For inelastic strains in a gross region surrounding a small crack, a plot of $\log a$ vs. N is linear for constant plastic strain range ($\Delta\epsilon_p$), or

$$\frac{da}{dN} = A_{11}(\Delta\epsilon_p)^r \, a \, , \tag{15}$$

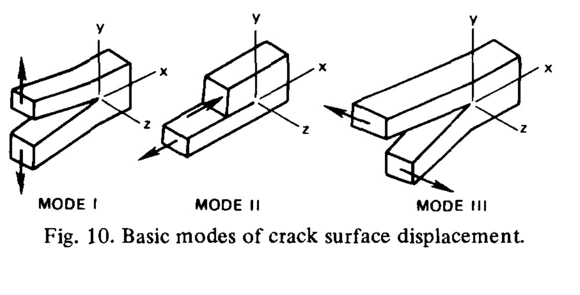


Fig. 10. Basic modes of crack surface displacement.

where A_{11} is a function of strain range and r is a constant. The driving force in this case, $(\Delta\epsilon_p)^r \, a$, is different from that derived from conventional fracture mechanics concepts which give the coupling $(\Delta\epsilon_p\sqrt{a})^\beta$.

In the case of low strains and large crack sizes, a parameter method developed by Carden [12] can be used to obtain linear relationships from mode I loading data. This method incorporates temperature (T), cycle frequency (ν), and R in the correlation, which is based on the expression

$$da/dN = MP^\alpha \, . \tag{16}$$

In this equation, M and α are constants, while P is a parameter given by

$$\begin{aligned} P = \{&[A_{1101} \exp(-\Delta H_1/R_1T) \\ &+ A_{1102} \exp(-\Delta H_2/R_1T)]\,(1/\nu)^p \\ &+ C_{1101} \exp(-\Delta H_3/R_1T)(1/\nu)\}(K_e^2 - K_{eth}^2)^s \, , \end{aligned} \tag{17}$$

where $A_{1101}, A_{1102}, C_{1101}, \Delta H_1, \Delta H_2, \Delta H_3, R_1, p$ and s are constants independent of temperature and frequency.

The term $A_{1101} \exp(-\Delta H_1/R_1T)$ accounts for low-temperature behavior over all frequency ranges and also includes thermal effects on strength, hardening, strain rate, and modulus. The term $A_{1102}\exp(-\Delta H_2/R_1T)(1/\nu)^p$ accounts for high-temperature, high-frequency effects. If the behavior is frequency independent, p is zero. Finally, $C_{1101} \exp(-\Delta H_3/R_1T)(1/\nu)$

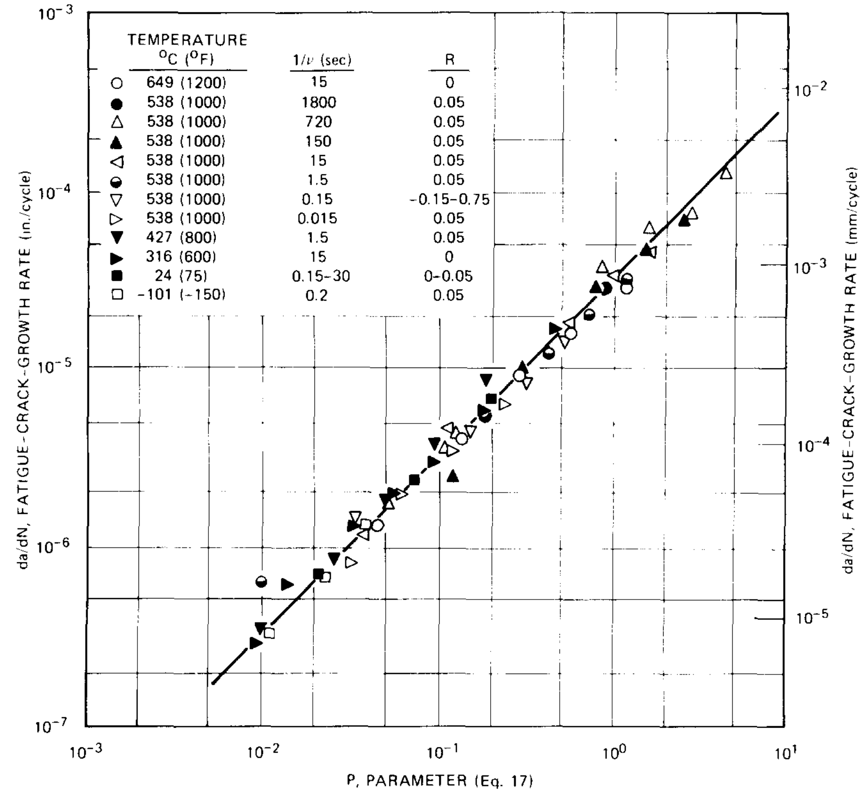


Fig. 11. Fatigue crack-growth rate as a function of parameter P [eq. (17)].

is time dependent and accounts for purely time-dependent processes. This term does not contribute significantly unless the frequency is low. The parameter does not specifically include wave shape and environmental dependencies.

The parameter method was used to treat data obtained by James [13–15] for type 304 stainless steel in air; the results are shown in fig. 11. The correlation is remarkable since (1) the data span over five decades of frequency [at 538°C (1000°F)]; (2) the temperatures span the range from –101°C (–150°F) to 649°C (1200°F); and (3) the R values span the range from –0.15 to +0.75. Hence, this example indicates that linear relationships can be derived for a wide range of conditions, thus greatly enhancing design analysis capability.

Up to this point crack-initiation and crack propagation studies have been discussed separately, which is an acceptable practice. Beyond this, it is instructive to consider the fatigue process as a whole and to examine the various phases from data on single specimens. This was done at ANL [16] through postmortem scanning electron microscope studies of failed low-cycle-fatigue specimens. These were 6.35-mm dia. (0.25-in.) hourglass specimens, and from the information obtained the investigators estimated the number of cycles to crack initiation (N_0) as well as the crack-growth behaviors. The results were obtained from type 304 stainless steel specimens tested in air at 593°C (1100°F), a strain range of 1%, and a strain rate of 4×10^{-3}/s. Cycles with tensile, compressive, and symmetrical hold times of 1 min were used in three cases, and continuous cycling was employed in two cases. The crack-initiation phase was assumed to terminate at a crack length of 1–3 grains. Crack-growth information was determined from the spacings of the striations induced, except in the case of the tensile hold specimen which had intergranular fracture and consequently no striations. The data obtained from these specimens are shown in fig. 12. Striation measurement was begun in

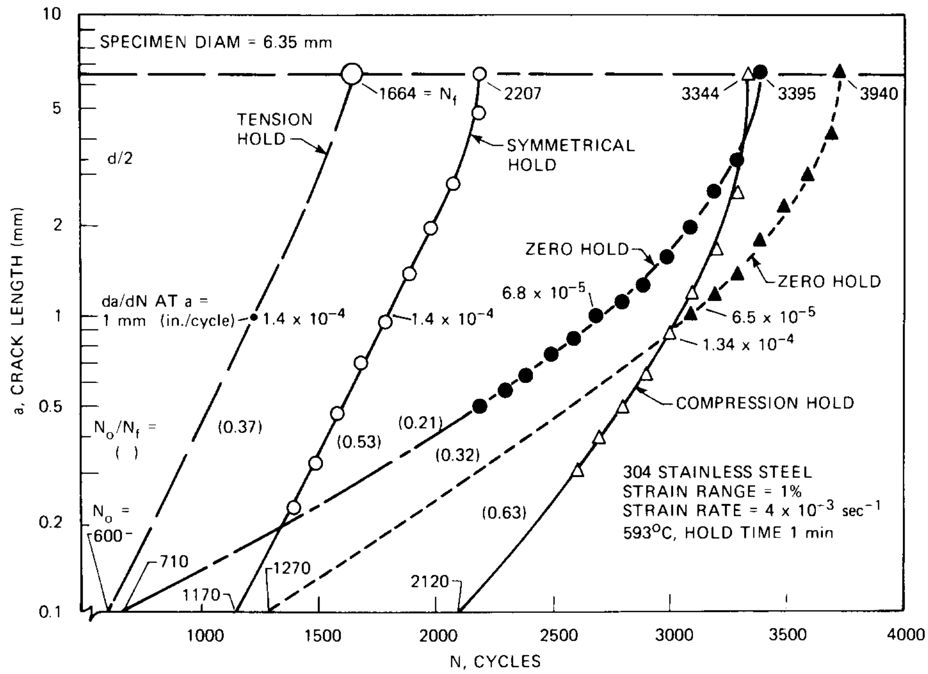


Fig. 12. Effect of hold condition on fatigue crack initiation and growth.

each case at a crack length of about 0.2 mm (0.008 in.) (~2 grain dimensions), which was taken as the demarcation between the crack-initiation and propagation phases. Note that in this case the crack length at the end of the initiation phase is much less than the 2.5-mm (0.1-in.) length referred to in discussing crack-initiation methods. For discussion, the crack growth rates for tensile hold were assumed to be equal to those for symmetrical hold, and a parallel curve was drawn.

Several observations can be made from these results concerning the ranking of low-cycle-fatigue failure, fatigue crack growth, and crack-initiation lives. With respect to low-cycle fatigue, the no-hold condition gives the greatest lives (N_f). However, the fatigue crack-growth periods account for major portions, and the crack-initiation period for no-hold is short in comparison to that for compressive hold. (The ratios, N_0/N_f, are shown in fig. 12.) The fatigue crack-growth curve for compressive hold is essentially parallel to that for symmetrical hold but displaced to the right, and the N_0/N_f ratios for both are much greater than for no-hold. Fatigue crack-growth rates (da/dN) at a = 1 mm (0.004 in.) are given in the figure for reference. On the basis of N_0, tensile hold would be expected to give the lowest life, although not by a large amount in comparison to lives for no-hold and symmetrical hold conditions. The latter give about equal lives. The compressive hold condition gave the longest crack-initiation period.

These results, with N_0 defined on the basis of a 0.2-mm crack length, indicate that failed low-cycle-fatigue specimens should be examined as described to adequately interpret the data obtained. To consider the fatigue failure process for these cases as being dominated by the crack-initiation phase, is clearly incorrect. On the other hand, the importance of the crack-initiation phase is demonstrated, and the influence of wave shape is seen to be dependent on the phase under consideration.

Studies such as the one just described have been used to determine the influence of surface roughness on N_0. Data obtained by Maiya [2] were shown in fig. 2; the trend is indicative of that observed from other tests. Increased surface roughness decreases the crack-initiation period, leading to the expectation that N_0 would be zero for low-cycle-fatigue specimens with surface scratches having depths of 1–2 grains. Wave shape and frequency effects have not been addressed in these examinations.

Fatigue crack-growth behavior continues to be the subject of intensive studies on an international scale. However, there are a number of areas where information is minimal or lacking. These include (1) fatigue crack-growth data for small cracks and large inelastic strains, especially creep strains; (2) crack-growth under mode III displacements; (3) wave-shape and hold-time effects; (4) environmental effects; (5) effects of overload and retardation; and (6) effects of temperature variations during straining.

The use of fatigue crack-growth methods requires that appropriate terminal, or "failure" conditions be specified. Potential candidates are (1) a maximum allowable growth rate [$\sim 1.0 \times 10^{-2}$ mm/cycle (4×10^{-4} in./cycle)]; (2) limits on crack length; (3) limiting K values; and (4) limits imposed by ratchetting or geometric instability conditions. The limits on crack length must be related to the specimen, or component, size and geometry. The limiting K value specified should account for the various factors involved, such as K_{Iscc}, for aggressive environments and modifications to allow for influences of the available energy in the system.

The fatigue crack-growth viewpoint offers an advantage in damage determination because present crack length is an obvious measure. From the preceding discussion it is seen that damage accumulation by this measure is a nonlinear function of cycle number. Damage is also associated with degradation of material properties such as lowering of K_{Ic}. The concepts developed for uniaxial, or essentially uniaxial, cases must be extended to cover multidimensional loadings in the generality observed in practice. In addition, the influences of neighboring cracks and/or external surfaces must be factored into the damage description and the method for analysis of cracked bodies.

5. Summary

A number of areas of agreement and unresolved issues were identified during the course of this study. The highlights of some of these are given in the following paragraphs.

The report was directed to the rather specific case of a structure at elevated temperature where the thermal and mechanical loadings are principally cyclic and where time-dependent processes such as creep-environmental reaction and metallurgical stability enter. Since here the inelastic strains are almost entirely cyclic and monotonic accumulation is considered to be relatively small, questions concerning monotonic creep or long periods of creep interspersed with short periods of cyclic strain were addressed only briefly.

It was emphasized by the authors throughout that the basic nature of fatigue damage is the initiation and growth of cracks, and an underlying tenet of the phenomenological approaches considered was the acceptability of extension of concepts from low-temperature behavior to high temperature. Basic variables were identified by the authors, but there were differences in treatment. The current approaches to fatigue behavior correlation and prediction are primarily for isothermal continuous-cycling response under uniaxial-loading and repeated cycle conditions. In fact, this statement reasonably describes general progress to date. However, this progress is not trivial; in view of the many variables involved, the correlation and prediction of repeated cycle behavior is a major accomplishment. On the other hand, much more is required to establish structural design procedures for general application, and the importance of this task warrants a broadened, carefully coordinated long-term attack in which the full range of applicable desciplines is brought to bear. To be included are microstructural studies, modeling (both at the microstructural and continuum levels), development of constitutive equations, and further implementation of the phenomenological approach.

Material stability and environmental effects are very important at elevated temperatures. Not only are they important individually, but the combined effects are synergistic. Inelastic straining causes metallurgical changes that manifest themselves in hardening, softening, strain aging, etc. conditions which, in turn, affect rheological behavior (inelastic stress–strain response) of the material. In addition, environmental influences plus time at elevated temperature affect the metallurgical changes which influence both rheological behavior and the capacity to sustain deformations.

Studies needed for advancement in fatigue behavior correlation and prediction relate to several areas. A primary need is the extension of the low-cycle-fatigue and fatigue crack-growth data bases. Low-cycle-fatigue tests involving long hold times or very low strain rates are needed to provide data on existence of saturation in cyclic life; on effects of wave shape, strain level, temperature, and environment; and on differences between materials. Strain aging, long-time embrittlement, and other metallurgical effects should be addressed in these studies.

Almost no results are available on fatigue crack growth for cycle periods longer than 30 min, and numerous unresolved questions have been raised concerning fatigue crack growth in the long time regime. To be examined are items such as the effects of mean stress, extent of applicability of linear elastic fracture mechanics methods, and the extent to which strain-based approaches can be used to correlate behavior. Also of particular importance are high-temperature cases involving very small cracks (in their incipient stages) and large inelastic strains (especially creep strains).

A second area in need of further investigation involves thermomechanical cycling effects on crack initiation and growth. Operating conditions for structures and components generally involve variations in temperature as well as stress and strain states. Since current approaches have been primarily developed for modeling isothermal continuous-cycling behavior, they do not contain a general repository or measure of damage contributions that result from constituent parts of changing cyclic conditions. Thus, development of data bases on thermomechanical cycling effects is essential to understanding and to modeling.

Thermomechanical effects need to be viewed differently depending upon whether the material is in the crack-initiation or crack-growth phase. The former involves a local event that can generally be treated by considering only local strains and temperatures, but larger regions of structures must be considered in connection with crack growth.

Closely tied to this and identified as an area of outstanding need concerns cumulative damage and embraces both development of precise definitions of damage and determinations of appropriate measures for modeling. The "damage" incurred by a material as a result of inelastic deformations history (in particular, cyclic inelastic straining history) is not totally understood, as the report indicates. It is known that inelastic strain is a major variable, and there is abundant evidence from tests under repeated uniaxial cycle conditions to support this contention. Use of inelastic strain as a correlating variable gives recognition to the fact that fatigue requires local reversed inelastic flow and that a correct parameter for *damage production* under repeated cycle conditions is the inelastic strain amplitude. It is also recognized that factors other than inelastic strains are important to damage production. These factors include temperature, environment, and metallurgical changes, and the synergistic effects produced when combined with inelastic strains.

Uniaxial testing under repeated cycle conditions has provided a large amount of information concerning damage production. However, to achieve needed advancements, greater understanding than can be derived in this way is required to address damage under nonrepeated cycle conditions or the influences of priot inelastic strains on fatigue and crack growth and vice versa. Thus, broader examinations to identify and study damage and measures of damage are essential. Microstructural examinations and detailed mechanistic studies can provide invaluable informations. In addition, multiaxial considerations must be factored in.

Characterization of behavior under time-varying multiaxial stress and strain states has not received attention commensurate with its importance. Because multiaxial stress states generally exist in structures and components in service, the attendant effects must be considered. This applies not only to influences of time-varying stress and strain states on fatigue life, but also to accounting for damage accumulation on a given plane at a given location. In the general case, correlating variables must possess both magnitude and direction to be fully appropriate, and the guidance needed for their specification cannot be acquired without detailed multiaxial experiments. The generality to be realized from the broader view advocated is essential to the development of improved structural design procedures.

Many other areas of importance are enumerated and discussed in the report [1], but they generally apply to more specific aspects. A well-planned long-range program should at a minimum address the areas described above.

Acknowledgements

The author gratefully acknowledges A.E. Carden, C.E. Pugh, and R.W. Swindeman for reviewing this paper and M.R. Sheldon for editing the manuscript.

References

[1] A.E. Carden, L.F. Coffin, Jr., S.S. Manson, L.K. Severud and W.L. Greenstreet, ORNL-5073 (Jan. 1977).
[2] P.S. Maiya, ORNL-5015 (June 1975) p. 21.
[3] L.F. Coffin, in: 1976 ASME-MPC Symposium on Creep–Fatigue Interaction, MPC 3, R.M. Curran, ed. (ASME, 1976) p. 349.
[4] S.S. Manson, G.R. Halford and M.H. Hirschberg, Proc. 1st Symp. on Design for Elevated Temperature Environment, San Francisco 10–12 May (1971) p. 12; Discussion, ASME (1971) p. 25.
[5] S.S. Manson, STP-520, 744-782, ASTM (1973).
[6] L.F. Coffin, Jr., Fracture, Proc. 2nd Int. Conf. on Fracture, Brighton, April (1969) p. 643.
[7] L.F. Coffin, Jr., Proc. Air Force Conf. on Fatigue and Fracture of Aircraft Structures and Materials, AFFDL-TR70-144 (1969) p. 301.
[8] (a) D.R. Diercks and D.T. Raske, ANL-76-95 (Dec. 1976); (b) unpublished ANL data.
[9] K. Kanazawa and S. Yoshida, Proc. Int. Conf. on Creep and Fatigue in Elevated Temperature Applications, Philadelphia, Sept. (1973).
[10] T. Udoguchi, Y. Asada and I. Ichino, Proc. Int. Conf. on Creep and Fatigue in Elevated Temperature Applications, Philadelphia, Sept. (1973).
[11] S.S. Manson, Case Western Reserve University (Apr. 1977).
[12] A.E. Carden, Proc. Int. Conf. on Creep and Fatigue in Elevated Temperature Applications, Philadelphia, Sept. (1973).
[13] L.A. James, Proc. 5th Nat. Symp. on Fracture Mechanics, 31 Aug.–2 Sept. 1971, Urbana, Part 1, ASTM-STP-513 (1972) p. 218.
[14] L.A. James, Nucl. Technol. 14 (1972) 163.
[15] L.A. James and E.B. Schwenk, Met. Trans. 2 (1971) 491.
[16] R.W. Weeks, D.R. Diercks and C.F. Cheng, ANL-8009 (Nov. 1973).